The Ballistic Pendulum: Finding a Hidden Speed

The Ballistic Pendulum · oPhysics

Senior Secondary (Years 11-12)	~30 minutes	Science · Physics

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Simulation: https://ophysics.com/e3.html
Curriculum links: momentum · energy transfer

Learning intentions

- I can apply conservation of momentum to the inelastic collision stage of a ballistic pendulum.
- I can apply conservation of energy to the pendulum-swing stage.
- I can combine both principles to calculate a projectile's initial speed from the pendulum's swing height.

Getting started

Open The Ballistic Pendulum and read the on-screen instructions. Note the mass of the projectile, the mass of the block, and record the maximum height the pendulum reaches after the projectile embeds in it.

Tasks

1. Using the swing height h, calculate the speed of the block-and-projectile immediately after the collision using $v = \sqrt{(2gh)}$.

Working:

2. Using conservation of momentum for the collision stage ($m \cdot v0 = (m + M) \cdot v$), calculate the projectile's original speed v0.

Working:

3. Reveal the simulation's actual initial velocity and compare it to your calculated answer. Calculate your percentage error.

4. Repeat the full process (Tasks 1–3) for two more combinations of projectile mass and block mass.

Projectile mass	Block mass	Swing height	Calculated v0

Extension (optional)

In a real ballistic pendulum experiment, why can't the bullet's speed simply be measured directly, and why does this two-stage method (momentum, then energy) work around that?